

The Role of the Nanoconfinement Effect in the Adsorption of Carbon Dioxide by Porous Liquids

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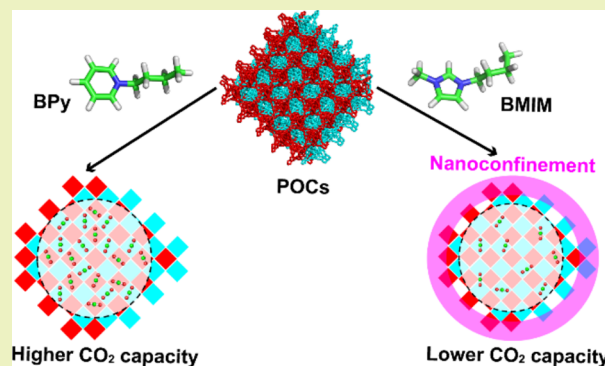


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ABSTRACT: Porous liquids based on porous organic cages (POCs) have attracted much attention recently due to their notable gas adsorption and separation ability. Compared with well-developed type II porous liquids, POC-based type III porous liquids were less explored and understood, especially those POCs solvated in ionic liquids. A previous experimental study showed that the POC cluster solvated in different ionic liquids presented different CO₂ capacities, while the mechanism behind the molecular level is unknown. The CO₂ diffusion mechanism needs to be further uncovered to guide solvent screening for the POC-based type III porous liquids. In this study, two porous liquid systems with the same POC cluster solvated in different ionic liquids were investigated through computational methods. The CO₂ capacity of porous liquids composed of the POC cluster and [BPy]⁺[NTf₂][−] (BPy_{NTf2}) is higher than that of the consisted POC cluster and [BMIM]⁺[NTf₂][−] (BMIM_{NTf2}), which is supported by the experimental results. The detailed dynamical and structural analyses demonstrated that the CO₂ diffusion rate was lowered by the interface between the solvent and the POC cluster in both systems. Moreover, in the BMIM_{NTf2} system, the ring part of cation molecules concentrated above the cage window efficiently hinders the CO₂ entering the POC cluster inside, which explains the difference in the CO₂ capacity in the two systems. The observed nanoconfinement effect at the interface between the solvent and the POC cluster surface challenged the assumption that chemical groups of solvent molecules entered inside cages, leading to a steric effect on the CO₂ diffusion dynamics. Our findings provide new insight into how the nanoconfined structure impacts the CO₂ adsorption dynamics in porous liquids. This may contribute to the future screening or designing of solvents for POC-based type III porous liquids.



KEYWORDS: type III porous liquids, molecular dynamics simulation, CO₂ capture, nanoconfinement, ionic liquids, organic cages

INTRODUCTION

As an emerging type of liquid material, porous liquids (PLs) have attracted much attention in the past decade due to their great ability of gas adsorption and separation.^{1–12} Porous liquids introduced fluid properties into the porous solid, combining the advantages of both porous solids and liquids.^{5–7,13–18} Porous solids such as porous organic cages (POCs),^{19–25} metal–organic frameworks (MOFs),^{26–29} or zeolitic imidazolate frameworks (ZIFs) provide permanent porosity with rich gas adsorption behaviors.^{30–33} Meanwhile, liquids have a great ability to dissipate heat during adsorption and regeneration processes.^{13,34} Also, liquids do not suffer from mechanical fatigue or physical aging, which is common in solid materials.^{35,36} As of today, four types of porous liquids have been applied to gas adsorption or separation.^{1–3,6,37} Type I PL is a neat liquid composed of one type of molecule that provides permanent intrinsic porosity.^{38–41} Type II PL combines the porous host and solvent, such as the porous organic cages solvated in the sterically hindered crown-ether liquids.^{42–44} Type III PL shares the same combination method

as that of type II PL, while the solutes in type III PL usually are nanoparticles instead of organic cages.^{7–11,15} Type IV PL is another single-component liquid with a lower melting point at room temperature.⁴⁵ Among them, type II and type III PLs attract much attention due to the large combination of the porous host and solvent, which provides potential tunable applications for gas capture.^{37,46}

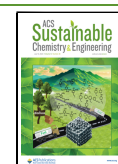
POCs are relatively new porous materials with multiple advantages such as highly specific surface areas, stable porosities, open windows, and tunable structures, so they have been widely tried in multiple areas including gas adsorption and separation, drug delivery, and energy storage and conversion.^{19,47–49} Due to their better solution dispersi-

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bility and processability compared with other extended porous frames, POCs have great potential to form type II porous liquids combined with the solvent in an appropriate size. For instance, an experimental study showed that imine-based cages could be solvated in 15-crown-5 with an extremely high concentration (44%) to provide stable porosity, and it can also form porous liquids with the hexachlorophene solvent which is too big to enter the cage inside.⁵ Wang et al. studied the gas selectivity of porous liquids composed of POCs and 15-crown-5 in different concentrations using computational methods.⁴² They found that the porous liquids absorbed CO₂ more efficiently than methane and nitrogen at all concentration levels. Li et al. further investigated the CO₂ adsorption capacity of porous liquids composed of POCs and ten different organic solvents.⁴³ They found that the CO₂ capacity is higher at low temperatures and high pressure. Sholl et al. applied a machine learning method to select appropriate solvents for POCs based on the pK_a values and several other features.⁵⁰ They predicted about 70 solvents for the POCs to form type II porous liquids. Although their predictions are inconsistent with experimental results, they proposed a feasible method to screen the solvent for POCs to form type II porous liquids.

To build the porous liquids, the solvent can not only be organic solvents but also ionic liquids.³ Ionic liquids (ILs) have great physicochemical features such as high structural tunability, ideal affinity with target molecules, and negligible vapor pressure of most systems.² They have been widely used to build type III porous liquids combined with extended porous frames such as MOFs,³¹ COFs,⁹ and ZIFs.¹⁵ Nenoff et al. argued that there are more than 10¹⁶ combinations of solute and solvent for porous liquids composed of ionic liquids.³⁷ In other words, if it consists of POCs and ionic liquids, the porous liquids may obtain higher tunability and diverse mechanisms of gas adsorption. Moreover, when participating in type III liquids, solid-state properties, such as chiral recognition, of POCs can be retained.^{1,52} A previous experimental study conducted by Greenaway et al. synthesized serials of stable POC-based type III porous liquids.⁵³ Among them, the system composed of POCs and the ionic liquid (BPy₂NTf₂) shows an obviously higher CO₂ capacity than the neat system of ionic liquids between 25 and 100 °C at 1 bar. The system composed of POCs and the ionic liquid (BMIM₂NTf₂) shows almost equal amounts of the CO₂ capacity to the neat ionic liquids. This interesting difference implies that solvent molecules may play a vital role in the dynamical mechanism of CO₂ adsorption and diffusion in POC-based porous liquids. Previous studies on the diffusion dynamics of water molecules into a nanoporous silica gel structure demonstrated that nanoconfined structures have the ability to limit the diffusion process of gas and ions.^{54,55} However, in the POC-based type III porous liquids, it is less understood how the nanoconfined structure at the interface between the solvent and the porous host affects the gas and solvent diffusion dynamics. It is reasonable to suggest that the diffusion mechanisms of gas entering the inside of the porous host in POC-based porous liquids are more complicated. This is because the complex surfaces of the POC cluster may lead to unexpected nanoconfinement effects with different solvent molecules. Iovnov et al. proposed that the incomplete packing of solvent molecules at the solute surface may unexpectedly change the gas diffusion dynamics.⁵⁶ However, the specific nanoconfined structure formed by the POC surface and the ionic liquid solvent is less explored currently, which is a major obstacle to

rationally screening or designing solvent molecules for the tunable, task-specific type III POC-based porous liquids.

In this study, two different type III POC-based porous liquid systems, CC3/BMIM₂NTf₂ and CC3/BPy₂NTf₂, were built up to uncover the dynamical mechanism of the CO₂ diffusion behavior in these two porous liquids using the MD simulation method. Each system was divided into four regions: (1) the cluster inside, (2) the surface layer, (3) the outer layer, and (4) the solvent phase to investigate the dynamic properties of gases and solvents. Root-mean-square fluctuation (RMSF), number density, radial distribution functions (RDF), and distance distribution were applied to elucidate how the nanoconfined structure at the interface between POCs and the solvent affects the CO₂ diffusing into the inside of the POC cluster, which determined the capacity of CO₂ in different solvent systems. The difference in the CO₂ capacity in two porous systems from our computational results is supported by the previous experimental results.⁵³ Molecular dynamics analysis showed that the nanoconfinement effect was more obvious in the BMIM₂NTf₂ system. It was limited at the interface between the surface of POCs and the solvents with a scale of ~1 nm. We noticed that the BMIM molecule's ring part accumulates above the organic cage window, making it difficult for CO₂ to diffuse into the porous cages. Our discovery challenges the idea that some solvent molecules obstruct CO₂ diffusion by entering the cage. We hope our findings can provide a new perspective on the CO₂ adsorption behavior in POC-based porous liquids. This could help in the future screening or designing of type III POC-based porous liquids.

METHODS

The general AMBER force field (GAFF) was used to simulate ionic liquids and porous organic cages, which has been proven to provide reasonable thermodynamic and transport properties of the ionic liquid system.^{57,58} The structures of ions and chiral cages were optimized at the Hartree–Fock (HF) level and the 6-31G(d,p) basis set, and then their partial charges were determined by the RESP method using Antechamber tools.^{59,60} All of the partial charges of atoms in the ionic liquids are scaled by a factor of 0.8.⁵⁷ The EPM2 force field simulated the CO₂ molecules as a linear rigid model.⁶¹ The C–O bond length is 1.149 Å with Q_c = 0.6512 and Q_o = −0.3256. The N₂ molecule was constructed by a three-site model, in which one virtue atom was located in the middle of two nitrogen atoms with a constrained distance of 1.1 Å.^{62,63} Partial charges of the virtue atom and the nitrogen atom are 0.964 and −0.482, respectively. Structural optimization and energy calculation were performed using the Gaussian16 package.⁶⁴

There are two steps for system modeling. Briefly, the first step is solvating the POC nanocluster into ionic liquids to form the porous liquid box and then performing MD simulation to obtain a stable porous liquid system. The second step is adding CO₂ and N₂ molecules with a ratio of 1:1 into both sides of the stable porous liquid system along the Z-axis. Then, MD simulation will be performed again to investigate the dynamic behavior of diffusion and the adsorption of CO₂ in porous liquids. The N₂ molecules were used to maintain the gas phase of the whole system during the adsorption of CO₂.

The POC cluster consisting of CC3-R and CC3-S cages in our system is constructed from the crystal structure of CC15-R/CC3-S (CCDC code: 1533787).⁶⁵ Each of the R and S cages in the crystal structure are aligned and replaced by optimized structures of single CC3-R or CC3-S, and the alignment guarantees the cages keep window-to-window packing mode. There are 118 cages in the cluster, with 62 CC3-R cages and 56 CC3-S cages (Figure 1A). Among these cages, 30 R cages and 25 S cages were exposed to the solvent. The nanocluster shows nearly a spherical shape with a complex surface, in

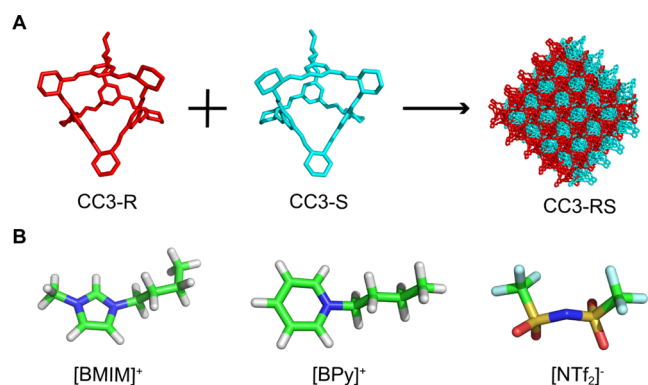


Figure 1. (A) Structure of the CC3-R (red) cage and the CC3-S (cyan) cage as well as the POC cluster. (B) Structures of cations and anions of the solvent. The carbon atom is green, the nitrogen atom is blue, the hydrogen atom is white, the oxygen atom is red, the sulfur atom is yellow, and the fluorine atom is cyan.

which the maximum distance between two cage centers is less than 80 Å (Figure S1). The POC cluster was solvated in different solvents using the PackMol package.^{66,67} The first solvent is BMIM_NTF₂ ion pairs and the other one is BPy_NTF₂ pairs (Figure 1B). For both porous liquid systems, there are 2400 pairs of cations and anions as the solvent, and the simulation box edges are at least 10 Å from the cluster surfaces.

10,000 steps energy minimization was conducted to the porous liquid system, and then the system was relaxed in 20 ns under the NPT ensemble ($P = 1$ bar, $T = 300$ K). The last trajectory configuration was selected for the second stage of system modeling. In the second stage, 2000 CO₂ and N₂ molecules were added to both sides of the box along the Z-axis (Figure 2A). For each simulation, 50 ns MD simulation under the NPT ensemble ($P = 1$ bar, $T = 300$ K) was performed, and the last 5 ns samples were used for the structural

and dynamical analyses. The integration time step was 1 fs and the neighbor list was updated every 10 steps. The cutoff of the van der Waals interactions was 12 Å. The short-range electrostatic interactions were cut off at 12 Å, and the particle-mesh Ewald method treated the long-range electrostatic interaction. All bonds were constrained using LINCS algorithms.^{68,69} The Nose–Hoover thermostat was used to control the temperature at 300 K and the C-rescale method was used to control the pressure at 1 bar. The MD simulations were performed using Gromacs 2022.06.^{70,71}

The lifetimes of gas and solvent molecules were calculated using the following equation⁷²

$$P(\tau) = \frac{1}{T} \sum_{t=1}^T \frac{N(t, t + \tau)}{N(t)} \quad (1)$$

where τ and N represent the lag time of sampling and molecule numbers, respectively. The diffusion coefficients were obtained from the slope of the mean square displacement (MSD), which was calculated by the Einstein relationship

$$D = \lim_{t \rightarrow \infty} \frac{1}{6t} \left[\frac{1}{N} \sum_{i=1}^N |r_i(t) - r_i(0)|^2 \right] \quad (2)$$

where r and N represent the molecule position and numbers, respectively.

RESULTS AND DISCUSSION

We first investigated the diffusion properties of different components in both porous liquid systems. The displacements of gas and ions were calculated along the Z-axis of the simulation box (Table 1). The diffusion coefficient of each component in the system was contributed by three areas: the gas phase, the solvent phase, and the cluster inside. The number ratio of the two gases is 1:1 in both systems. In the system solvated by BMIM_NTF₂, the diffusion coefficient of

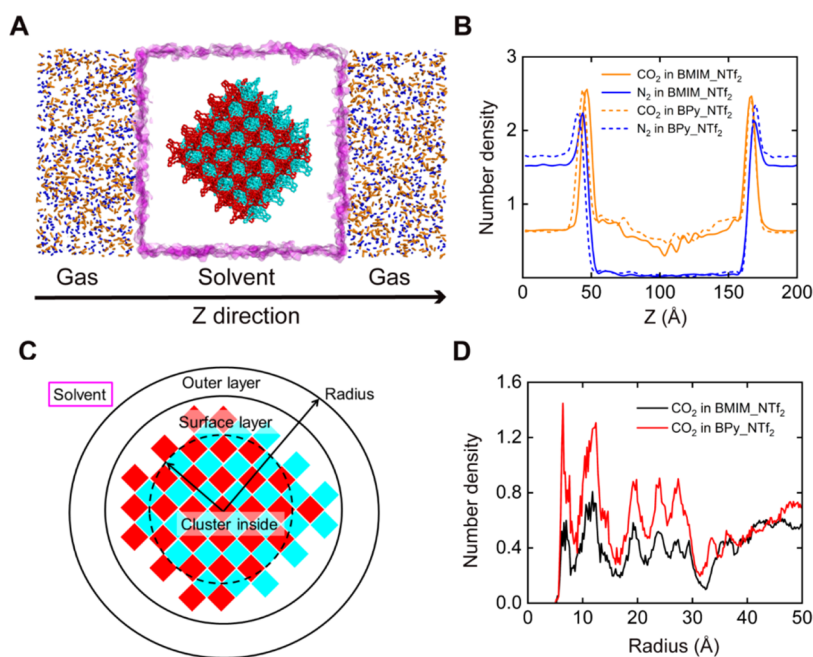


Figure 2. Investigation of the gas density in the porous liquids. (A) Diagram of the initial configuration of porous liquids for both systems. The porous cages (red and cyan) were solvated in the ionic liquids (magenta), which only showed the edges. The CO₂ (orange) and N₂ (blue) molecules were added to both sides of the solvent along the Z-axis. (B) Gas density distribution along the Z-axis when systems reached equilibrium. A solid line represents gas in the BMIM_NTF₂ system, while a dashed line represents gas in the BPy_NTF₂ system. (C) Areas around the POC cluster divided into multiple layers along the R direction. (D) CO₂ density distribution along the R direction in both systems. BMIM_NTF₂ is black, and BPy_NTF₂ is red.

Table 1. Diffusion Coefficients of Different Components for Two Porous Liquid Systems ($10^{-5} \text{ cm}^2/\text{s}$)

	CO ₂	N ₂	cation	anion
BMIM_NTF ₂	39.072 ± 0.999	108.131 ± 3.061	0.024 ± 0.004	0.019 ± 0.003
BPy_NTF ₂	33.688 ± 0.181	106.674 ± 2.856	0.022 ± 0.010	0.015 ± 0.010

gas components is much higher than those of cations and anions. Meanwhile, the diffusion coefficient of CO₂ is much lower than that of N₂. In the system solvated by BPy_NTF₂, the diffusion coefficient of different components followed the same order, that is, N₂ > CO₂ > ions. The diffusion coefficient of ions in these two systems is not significantly different. CO₂ molecules solvated in ionic liquids more easily than N₂, most of which remained in the gas phase, leading to a higher diffusion coefficient (Figure 2B). The observed gas diffusion coefficient demonstrates that ionic liquid solvents have great affinity to CO₂ instead of N₂, consistent with previous research.⁴²

We further investigated the position distribution of gas components for both systems. After the gas molecules have sufficiently diffused, the distribution of the gas density is along the Z-axis (Figure 2B). The CO₂ number density in the gas phase is obviously lower than the N₂ number density, while it is much higher than the N₂ density in the liquid phase. The number density of gas at the gas–solvent interface is higher than that of pure gas or the solvent phase, which is also observed in previous studies.^{42,73} Of the carbon dioxide molecules entering the solvent, the density of carbon dioxide in the BPy_NTF₂ system is slightly higher than in the BMIM_NTF₂ system, especially in the POC cluster region.

It is difficult to accurately examine the distribution of carbon dioxide inside and outside the POC cluster by virtue of the gas density distribution in the Z-direction alone since the ionic liquid surrounds the cluster in three directions. Therefore, we use a spherical shell model to investigate the distribution of CO₂ inside and outside the POC cluster (Figure 2C). The regions inside and outside the cluster are divided into three spherical shells on a nanoscale based on the cluster size (~80 Å): the cluster inside (radius <30 Å), the surface layer (30 Å < radius <40 Å), and the outer layer (40 Å < radius <50 Å). The areas of 50 Å < radius <60 Å were defined as the pure solvent. The distribution of the CO₂ number density was along with the radius R for both systems (Figure 2D). The number density of CO₂ inside the cluster in the BPy_NTF₂ system is much higher than that in the BMIM_NTF₂ system. Interestingly, the CO₂ density reduced rapidly at the surface layer of porous liquids in both systems and gradually increased along with the radius R. The number ratio of CO₂ inside the cluster of these two systems is 1.7 (BMIM_NTF₂: 43.878 ± 3.120; BPy_NTF₂: 74.479 ± 5.096).

Based on the above results, we found that it is harder for CO₂ to escape from the porous liquids than N₂. After the CO₂ molecules enter the solvent, the distribution along the R direction is not uniform. The BPy_NTF₂ system indeed captured the CO₂ more efficiently. Due to both systems sharing the same cluster structures, it is reasonable to speculate that the interface between the POC cluster and the solvent plays an important role in the CO₂ diffusion dynamics.

Diffusion coefficients can only reflect the average motion rate of CO₂ throughout the simulation box. To obtain specific dynamic details of CO₂ diffusion in the porous liquids, we calculated the survival probability of CO₂ molecules in different areas using eq 1 (Figure 3). For CO₂ away from

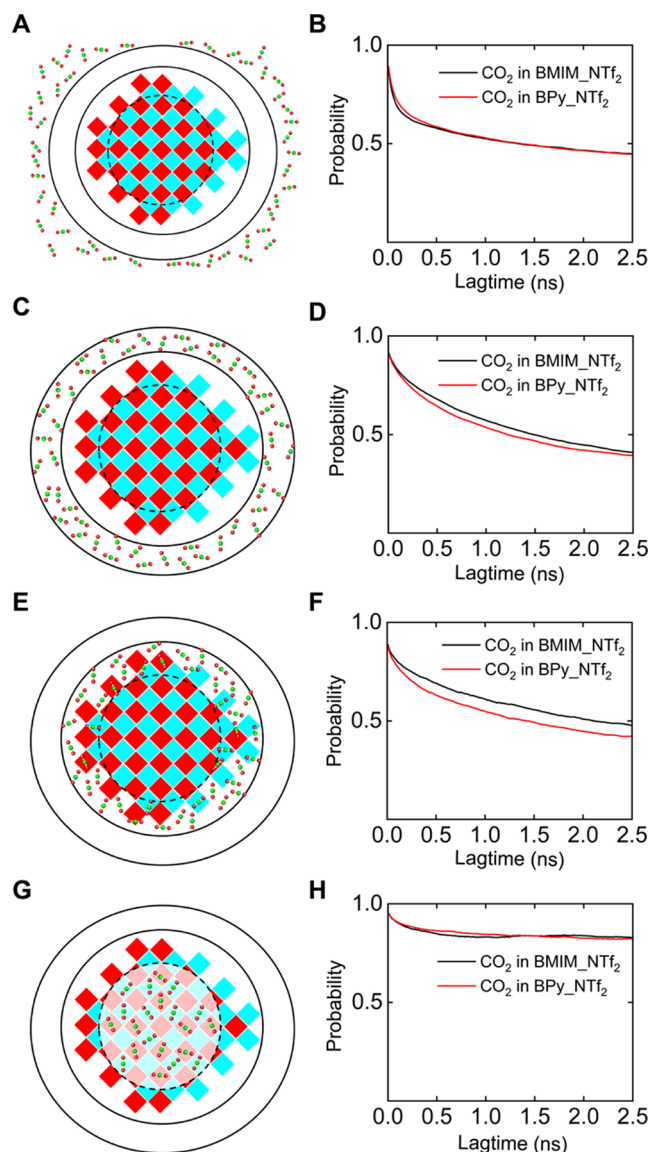


Figure 3. Investigation of survival probabilities of CO₂ molecules in different areas of porous liquids. (A) Diagram of CO₂ in the area away from the POC cluster. (B) Survival probability of CO₂ in the area corresponding to panel (A) for both systems. (C) Diagram of CO₂ in the outer layer of porous liquids. (D) Survival probability of CO₂ in the area corresponding to panel (C) for both systems. (E) Diagram of CO₂ in the surface layer of porous liquids. (F) Survival probability of CO₂ in the area corresponding to panel (E) for both systems. (G) Diagram of CO₂ inside the POC cluster of porous liquids. (H) Survival probability of CO₂ in the area corresponding to panel (G) for both systems.

the nanocluster (Figure 3A), their survival probability decayed rapidly at 300 ps in both systems (Figure 3B). While for CO₂ in the “outer layer” (Figure 3C), the survival probability decayed slower than that for CO₂ away from the nanocluster (Figure 3A,3D), which is also observed in the surface layer (Figure 3E,3F). Notably, the CO₂ molecules at the surface

layer of the BMIM-NTf₂ system show a higher survival probability than that of the BPy-NTf₂ system (Figure 3F). When entering the cluster inside, over 80% of CO₂ molecules remained in the cluster inside (Figure 3G,3H) for both systems. We also investigated the survival probability of cations and anions in different regions of porous liquids, as shown in Figure S2. Cations and anions in both systems kept over 80% survival probability in different shells around the POC cluster.

Compared to the solvent molecules, the diffusion behavior of CO₂ molecules in different shells is more complicated. When approaching the surface of the POC cluster, CO₂ molecules in both systems show a higher survival probability, demonstrating that the diffusion of CO₂ molecules was speeding down. The survival probability of CO₂ at the surface layer in the BMIM-NTf₂ system is higher than that in the BPy-NTf₂ system (Figure 3F), implying that different ion components may introduce distinct confined structures at the interface between the solvent and the POC cluster, leading to a diverse gas diffusion behavior.

To determine the relationship between ions and the potential confined structure at the interface, we first calculated the root-mean-square fluctuation (RMSF) of different parts of ions in different shells around the POC cluster (Figure 4). The

ions were selected from those retained in the original shell layer during the simulation. Two atoms at the ends of the cation were selected to compute the RMSF (Figure 4A,4C). The carbon atom of methyl on the ring of BMIM was selected and named the “head” atom, and the carbon atom of the chain terminal was selected and named the “tail” atom. For the cation BPy, the carbon atom on the ring and the carbon atom on the chain terminal were selected as “head” and “tail” atoms, respectively. RMSF values for both atoms gradually reduced when the cation approached the surface of the POC cluster. The RMSF of the cation in the solvent is greater than 4 Å, while it reduced to about 1 Å in the surface layer (Figure 4B,D). The RMSF value of the “head” atom shares the same scale with the “tail” atom of the cation. The center nitrogen atom was selected for the anion to compute the RMSF (Figure 4E). Like the cation, the RMSF of the anion in both systems also gradually reduced when it closed to the surface layer of the POC cluster.

The changes in RMSF values for the cation and anion demonstrate that the nanoconfinement effect is more obvious at the surface layer in porous liquids. However, the specific structural feature of the confined structure and its influence on CO₂ diffusion dynamics are unknown. A previous study demonstrated that the nanoconfinement effect is highly connected to incomplete solvents.⁵⁶ To investigate the change of the solvent structure at the interface, we further calculated the radial distribution function (RDF) of different parts of ions at the surface layer and the outer layer (Figure 5). The top panel shows the RDFs of cation–cation, cation–anion, and anion–anion at the surface layer (Figure 5A–5C), while the bottom panel is for the outer layer (Figure 5D–5F). The nitrogen atom on the ring of the cation and the nitrogen atom on the anion were selected to compute the $g(r)$. In the RDF of cation–cation for both systems, the cation number in the first cation shell around one cation is 1.8 at the surface layer (Figure 5A), while this number was raised to 2.3 at the outer layer (Figure 5D). The RDF of cation–anion at the surface layer in these two systems (Figure 5B) shows similar curves to that of the outer layer (Figure 5E). Besides, the second shell in the RDF of cation–anion is close to the first shell. We also checked the RDF of anion–anion in both systems and found that they share the same first-shell position at the surface layer and the outer layer (Figure 5C,5F).

The above RDF results of cations and anions in different layers demonstrate that the ion pair structure is kept intact even when approaching the POC cluster's surface. Based on the RDF results, we noticed that the density of ions decreased as they approached the cluster surface, so we compared the number density of the cation and anion for both systems at different layers, as shown in Figure 3S. The solvent composed of cations and anions was represented by a magenta shadow, as shown in Figure 3SA,3SB. In the outer layer, the number density of cations in both systems is about 1.7 nm⁻³, while it is reduced to 1.0 nm⁻³ at the surface layer. The anion also showed the same downward trend. It should be noted that the number density of cations is almost equal to that of anions.

Although the number density reduction of ions happened in the surface layer, both systems share the same scale of density reduction of ions. Meanwhile, the RDF of ions in these two systems also shows high consistency. Thus, we speculated that the difference in gas diffusion dynamics in these two systems was caused by other confined structures instead of the interaction change between cations and anions.

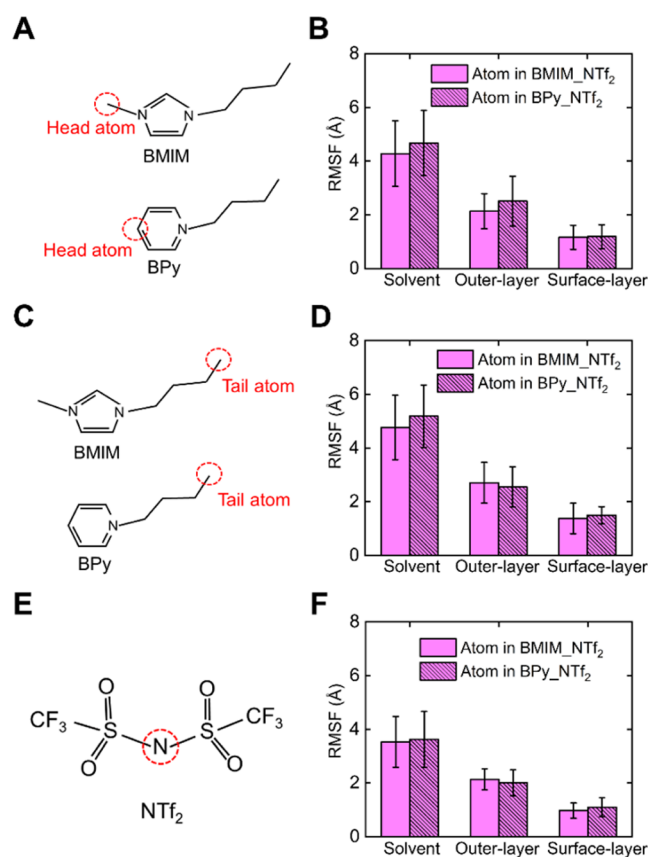


Figure 4. Root-mean-square fluctuation (RMSF) of different parts of solvent molecules. (A) The carbon atom of methyl of BMIM and the carbon atom on the ring of BPy selected to be the “head” atoms. (B) RMSF of the cation head atom in different areas of porous liquids. (C) Carbon atoms at the chain terminal of BMIM and BPy selected as the “tail” atoms. (D) RMSF of the cation tail atom in different areas of porous liquids. (E) The nitrogen atom in NTf₂ was selected to calculate the RMSF. (F) RMSF of the anion in different areas of porous liquids.

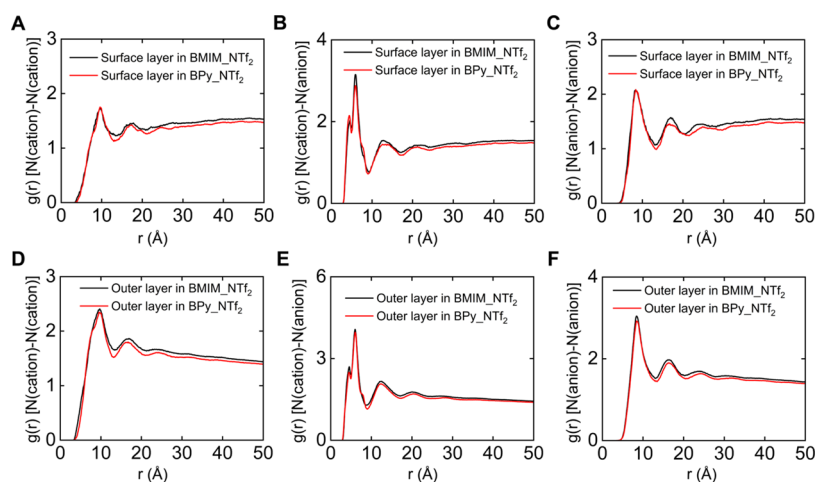


Figure 5. Radial distribution function (RDF) of ions. (A–C) RDF of cation–cation, cation–anion, and anion–anion in the surface layer, respectively. The nitrogen atoms in the cation and anion were used to calculate the RDF. (D–F) RDF in the outer layer following the same calculation method used in the surface layer.

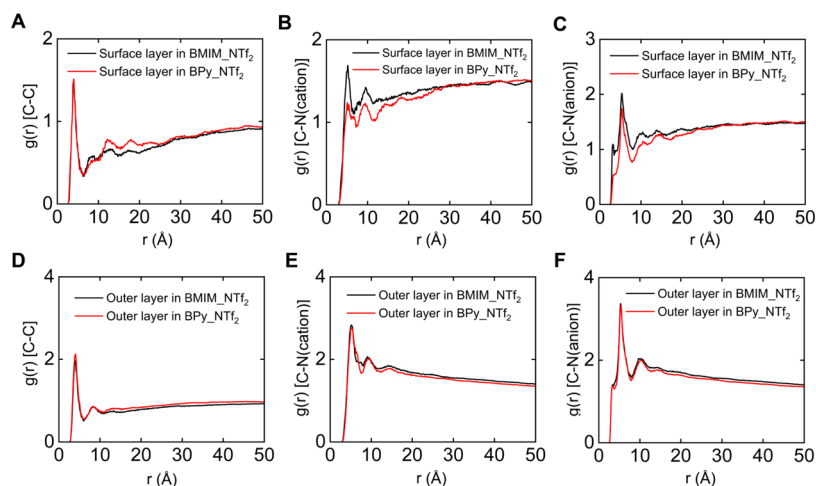


Figure 6. RDF results of CO₂ in the surface layer and outer layer. (A–C) RDF of CO₂–CO₂, CO₂–cation, and CO₂–anion in the surface layer, respectively. The carbon atoms in CO₂ and the nitrogen atoms in the cation were used to calculate the RDF. (D–F) RDF in the outer layer following the same calculation method used in the surface layer.

Based on the obvious difference in the CO₂ capacity between these two systems, the nanoconfinement effect should impact the distribution of CO₂ around the POC cluster. We calculated the RDF of CO₂–CO₂, CO₂–cation, and CO₂–anion at different layers (Figure 6). RDF results of CO₂–CO₂ show the same first-shell position at the surface and outer layer in these two systems (Figure 6A,6D). However, the RDF of CO₂–cation in the BMIM_NTF₂ system differs from that of the BPy_NTF₂ system at the surface layer (Figure 6B). Although the first-shell positions are the same for both systems, the number density of cations around CO₂ in the BMIM_NTF₂ system is obviously higher than that of the BPy_NTF₂ system at the surface layer. The difference in the number density of cations around CO₂ disappeared in the outer layer (Figure 6E). For the anion around CO₂, the first-shell position is closer to the CO₂ molecule in the BMIM_NTF₂ system than that in the BPy_NTF₂ system at the surface layer (Figure 6C). At the outer layer, the first anion shell around CO₂ in these two systems shares the same position (Figure 6F). Besides, the number of ions, including

cations and anions, in the first shell around CO₂ increased from the surface to the outer layer in both systems.

The RDF of CO₂–CO₂ distribution shows minor differences in BMIM_NTF₂ and BPy_NTF₂ systems. It implies that CO₂ molecules are not essential for confined structures. We have already excluded the assumption that the structural change of cation–anion pairs affects the gas diffusion dynamics. Thus, the CO₂ molecules are likely hindered by confined structures constructed by ions and the surface of the POC cluster. The RDF of CO₂–ions (cation, anion) demonstrates that ions at the surface layer play an important role in forming confined structures. Based on the previous discussion, we have shown that the number density of ions presented minor differences at the surface layer in both systems, and so the orientation or spatial configuration of ions at the POC cluster surface may have a great impact on the gas diffusion dynamics and thus the gas adsorption ability of porous liquids.

To uncover the confined structures formed by the solvent and cages at the cluster surface, we investigated the ion distribution around each CC3-R and CC3-S cage at the cluster surface. There are 30 CC3-R cages and 25 CC3-S cages

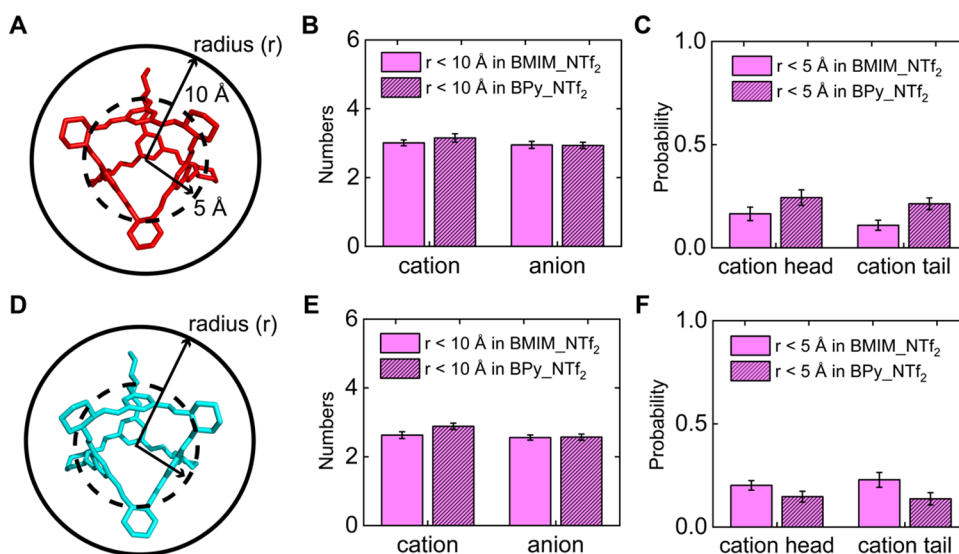


Figure 7. Investigation of the number density of ion molecules around the CC3-R and CC3-S cages at the surface of the POC cluster. (A) Diagram of the area in the CC3-R cage used to calculate the ion density. (B) Density of ions within r of 10 Å of CC3-R cages. (C) Proportion of ions within r of 5 Å to the ions within 10 Å for CC3-R cages. (D) Diagram of the area in the CC3-S cage used to calculate the ion density. (E) Density of ions within r of 10 Å of CC3-S cages. (F) Proportion of ions within r of 5 Å to the ions within 10 Å for CC3-S cages.

exposed to the solvent. We first calculated the average numbers of cations and anions around the CC3-R and CC3-S cages (Figure 7A,7D). Cations and anions within 10 Å of each cage center were used to compute the average value. We found that the numbers of ions are surprisingly consistent in BMIM-NTf₂ and BPy-NTf₂ (Figure 7B,7E). A previous study proposed that part of the cation molecule may have a chance to enter the cages inside at the surface of the porous host to hinder CO₂ diffusion.⁵³ Thus, we also checked the proportion of the number of cations entering the cage inside (<5 Å) to the total number within 10 Å (Figure 7C,7F). The ring part of cations was represented as the “head” atom in our study (Figure 4A), and the chain terminal of cations was represented as the “tail” atom (Figure 4B). For the CC3-R cages, the ratio in the BPy-NTf₂ system is slightly higher than the proportion in the BMIM-NTf₂ system (Figure 7C), while the trend reversed in the CC3-S cages (Figure 7F). Notably, the proportion is less than 0.5 for both types of cages in the two systems.

When the ring part of the cation enters the cage inside, it has a chance to cause a steric effect on the CO₂ diffusion dynamics. Thus, we checked the distance distribution of the “head” atom of the cation inside cages (Figure 8). For the CC3-R cages, the head atom of the cation in the BPy-NTf₂ system enters deeper inside the cage than that in the BMIM-NTf₂ system (Figure 8B). A similar distribution also happened in the CC3-S cages (Figure 8D).

The distance distribution of cations inside cages demonstrated that the two kinds of cations, BMIM and BPy, are allowed to enter the inside cages at the POC cluster surface. The ring of BPy is even closer to the cage center than the ring of BMIM, which may lead to a more obvious steric effect on CO₂ diffusion. However, the diffusion of CO₂ entering the POC cluster is not hindered in the BPy-NTf₂ system as in the BMIM-NTf₂ system. Thus, ions or part of ions entering the cage inside may not be the reason for the CO₂ capacity difference in these two systems. In other words, the confined structure formed outside the cages instead of inside, which challenged the assumption that the methyl group entering the cage inside hindered the CO₂ diffusion.

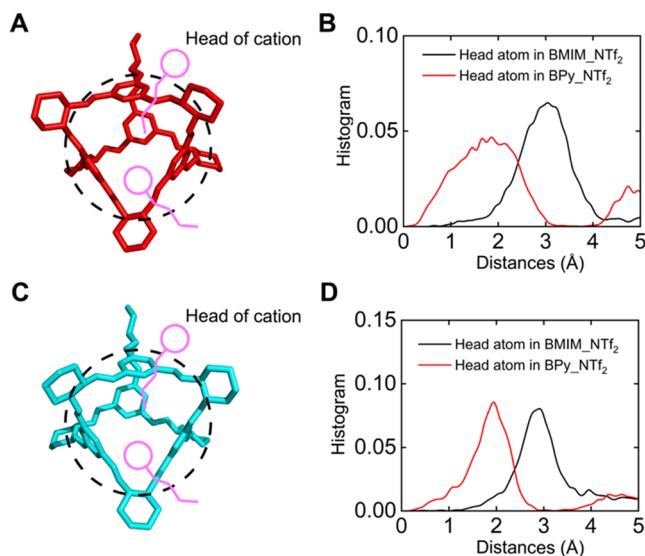


Figure 8. Distribution of the distance from the cation atom to the cage center. (A) Diagram of the CC3-R (red) cage and the cation (magenta) in which a circle represents the ring part (head). (B) Distribution of the distance from the cation head to the center of CC3-R cages for both systems. (C) Diagram of the CC3-S (cyan) cage and the cation (magenta). (D) Distribution of the distance from the cation head to the center of CC3-S cages for both systems.

Due to the ability of the cation ring to generate a steric effect, we calculated the number density of the cation ring outside of the cages (Figure 9). The distance to the cage center ranged from 5 to 10 Å for chiral cages. For the CC3-R cages, the cation ring of BMIM is more concentrated at ~8 Å from the cage center, while the BPy system presented a wider distribution with a lower number density (Figure 9B). For the CC3-S cage (Figure 9C), the ring of BMIM concentrated at 7.8 Å, while the density of the BPy ring has a wider distribution (Figure 9D).

The number density of the BMIM ring near the cage surface is more concentrated than that of the BPy ring at the cage

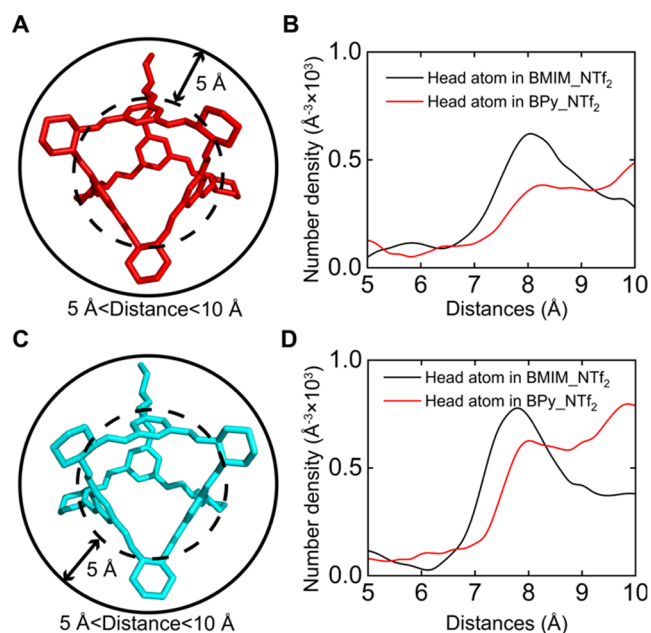


Figure 9. Investigation of the atom density at the surface of CC3-R and CC3-S cages at the surface of the POC cluster. (A) Definition of the CC3-R surface area in which the distance from the cage center is greater than 5 Å and lesser than 10 Å. (B) Head atom density of the cation at the surface of the CC3-R cage. (C) Definition of the CC3-S surface area. (D) Head atom density of the cation at the surface of the CC3-S cage.

surface. This demonstrates that the BMIM orientation has been changed compared with the BPy orientation at the cage surface. Consequently, BMIM rings concentrated at the cage windows hindered the CO₂ diffusing into the cluster inside. It should be noted that this nanoconfinement effect happened in a very limited area with a scale of ~1 nm.

To figure out how cations concentrated above the cage window affect the CO₂ molecules diffusing inside the cage, we investigated the angle distribution of cations and CO₂ relative to the window plane (Figure 10). Specifically, the angle was determined by two vectors. The first vector was defined by the head atom and center of the window plane, and the second vector was the normal vector of the window plane (Figure 10A). The window plane was defined by three nitrogen atoms of the window frame. The cation and CO₂ molecules with a distance from the cage center ranging from 6 to 10 Å were selected to calculate the angle distribution. The main peak of the angle distribution for the BMIM ring ranges from 30 to 45°, while there are two peaks for the angle distribution of the BPy ring (Figure 10B). It implies that more BMIM rings tend to be concentrated closer to the center of the window plane than the BPy rings. The angle distribution for CO₂ demonstrated that more CO₂ is concentrated above the center of the window plane in the BPy_NTF₂ system (Figure 10C).

The above angle distribution results demonstrated that BMIM rings tend to concentrate above the center of the cage window, which more efficiently occupies the limited space of the complex surface of POCs. This confined structure lowers the chance of CO₂ molecules entering the cage inside through the center area of the cage window. On the contrary, in the BPy_NTF₂ system, CO₂ molecules can diffuse into the cage inside through the cage window's center area due to the BPy ring's wide angle distribution. Thus, the confined structure

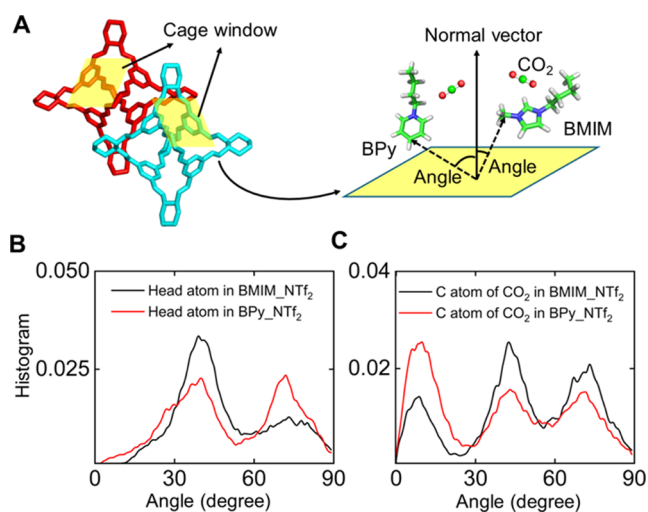


Figure 10. Angle distribution of the cation and CO₂ relative to the cage window. (A) Diagram of angle distribution. For simplicity, BMIM and BPy are shown in the same figure. (B) Angle distribution of the head atom of BMIM and BPy in two systems. (C) Angle distribution of the carbon atom of CO₂ in two systems.

above the cage window in the BMIM_NTF₂ system limited the rate of CO₂ molecules diffusing into the cages at the POC surface.

We also checked the number density of CO₂ entering the cages at the POC cluster surface (Figure 11). In the BPy_NTF₂

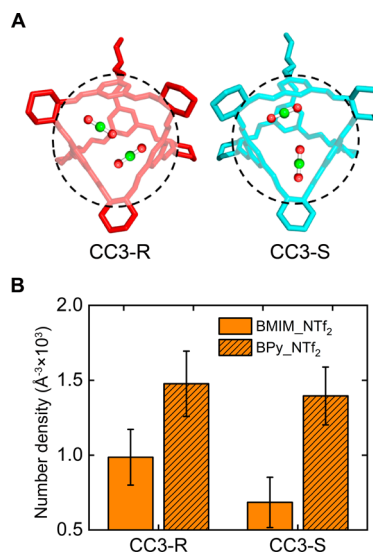


Figure 11. Investigation of the CO₂ density inside CC3-R and CC3-S cages at the surface of the POC cluster. (A) Diagram of CO₂ coordinated in both types of cages. (B) Number density of CO₂ in both systems.

system, the densities are 1.476 ± 0.218 and 1.395 ± 0.193 nm⁻³, corresponding to CC3-R and CC3-S cages. In the BMIM_NTF₂ systems, the densities are 0.986 ± 0.186 and 0.684 ± 0.168 nm⁻³, respectively. It is easier for CO₂ to go through the interface between the solvent and the POC cluster in the BPy_NTF₂ system. Thus, the CO₂ density inside the cages at the cluster surface provides a solid proof of the confined structure in the BMIM_NTF₂ system hindering the CO₂ diffusion dynamics.

Based on the above analysis, it is reasonable to propose that the lower CO₂ uptake ability in the BMIM_NTF₂ system is kinetic in origin. In the BMIM_NTF₂ system, the number of pathways of CO₂ molecules diffusing into the cage inside is limited due to the nanoconfinement effect. If CO₂ molecules go through the center area of the cage window, which is the main path for gas diffusion, they must overcome a higher barrier due to the confined structure above the cage window. In the BPy_NTF₂ system, the diffusion pathway is diverse due to the wide distribution of the angle of the cation ring relative to the cage window (Figure 10B). From the molecular perspective, the CO₂ molecule in the BPy_NTF₂ system makes it easier to enter the surface cage than in the BMIM_NTF₂ system (Figure 11B). We checked the CO₂ adsorption rate by the POCs on the nanosecond scale (Figure S4). We found that the BPy_NTF₂ system showed higher rates than the BMIM_NTF₂ system in different temperatures and pressures. Thus, the kinetic difference caused by confined structures is the main reason for the CO₂ capacity difference between the two systems.

It should be noted that the CO₂ capacity difference for the two systems in our research is not completely aligned with the experimental study.⁵³ The experimental work showed that the BMIM_NTF₂ system barely absorbs extra CO₂ after adding the POCs into the solvent compared with the BPy_NTF₂ system. In our study, although the BMIM_NTF₂ system showed a lower CO₂ capacity than the BPy_NTF₂ system, it still captured a small amount of gas. We proposed that there are two reasons for the unaligned results. As discussed above, CO₂ entering the POCs inside is kinetic in origin due to the nanoconfinement effect. Some CO₂ molecules can still overcome the barrier and diffuse into the surface cage in the BMIM_NTF₂ system. The other reason is that the modeling size in our simulation is different from that in the experimental research. The radius of POCs in the experiment reached the order of several microns to tens of microns, while the nanocluster is ~8 nm in our study. As the cluster radius increases, the volume-to-surface area ratio also increases. The nanocluster in our modeling can be considered a part of the “surface” of the real cluster used in the experimental research, leading to an overestimation of the CO₂ number ratio inside the POCs. Nevertheless, our findings still support the conclusion that confined structures in POC-based porous liquids play a vital role in the CO₂ capacity.

CONCLUSIONS

In summary, we investigated the dynamical behavior of CO₂ diffusing into POC-based type III porous liquids using the all-atom molecular dynamics simulation method. Structural and dynamical analyses showed that the CC3-R/S cage cluster solvated in BPy_NTF₂ ionic liquids presented a higher CO₂ uptake ability than the cluster solvated in BMIM_NTF₂, which is supported by the experimental results.⁵³ The CO₂ diffusion rate was reduced near the surface of the POC cluster, and this deceleration effect is more obvious in the BMIM_NTF₂ system. We found that the nanoconfined structure at the interface between the solvent and the POC cluster in the BMIM_NTF₂ system determined the efficiency of the CO₂ diffusion process into the inside of the POC cluster. Specifically, compared with the ring part of the BPy molecule, the BMIM ring tends to concentrate above the window of the cage. Although this structural feature of the ring does not change the density of cations at the whole surface layer, it hindered the CO₂ molecules entering the POC cluster in the BMIM_NTF₂

system. Consequently, the BPy_NTF₂ porous liquid system presented a higher CO₂ uptake ability than the BMIM_NTF₂ system. In our study, the nanoconfinement effect has a limited length scale with ~1 nm. Further studies are required to uncover the relationship between the size of the solvent molecule and the range of the nanoconfinement effect for the tunable solvent design of type III porous liquids.

Both CC3-R and CC3-S cages have the same impact on the solvent molecules in the surface layer. The number density and orientation of ions around the CC3-R cage presented the same features as those around the CC3-S cage, leading to the same CO₂ diffusion behavior and gas uptake ability. Thus, the chiral cages caused the same nanoconfinement effect in the porous liquids, consistent with the experimental results that CC3-R/CC3-S porous liquids show a similar CO₂ uptake ability with CC3-S porous liquids.⁵³ The chirality of the porous host may not be an essential factor affecting the gas diffusion dynamics. This finding may accelerate further screening and design of ionic liquids for porous liquids.

Overall, the porous liquids composed of organic cages such as CC3-R/S and ionic liquids have a high potential to capture CO₂. Our research proposed that the CO₂ capture efficiency is governed by the nanoconfinement effect at the interface between the solvent and the solute surface in the POC-based type III porous liquids. Our study provides atomic insight into the dynamics of CO₂ diffusing into porous liquids with different solvents. Further experimental and theoretical studies should be performed to uncover the role of the complicated interface between the solvent and the porous host on the gas diffusion dynamics for the future design of type III porous liquids.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acssuschemeng.4c02556>.

Distribution of the distance between cages, survival probability of ions in different areas, the number density of ions around the nanocluster, and the CO₂ adsorption rate by the POCs in different temperatures and pressures (PDF)

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Notes

The authors declare no competing financial interest.

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REFERENCES

- (1) Wang, D.; Xin, Y.; Yao, D.; Li, X.; Ning, H.; Zhang, H.; Wang, Y.; Ju, X.; He, Z.; Yang, Z.; Fan, W.; Li, P.; Zheng, Y. Shining Light on Porous Liquids: From Fundamentals to Syntheses, Applications and Future Challenges. *Adv. Funct. Mater.* **2022**, 32, No. 2104262.
- (2) Egleston, B. D.; Mroz, A.; Jelfs, K. E.; Greenaway, R. L. Porous liquids - the future is looking emptier. *Chem. Sci.* **2022**, 13, 5042–5054.
- (3) Li, X.; Chen, K.; Guo, R.; Wei, Z. Ionic Liquids Functionalized MOFs for Adsorption. *Chem. Rev.* **2023**, 123, 10432–10467.
- (4) Zou, Y.-H.; Huang, Y.-B.; Si, D.-H.; Yin, Q.; Wu, Q.-J.; Weng, Z.; Cao, R. Porous Metal-Organic Framework Liquids for Enhanced CO₂ Adsorption and Catalytic Conversion. *Angew. Chem.-Int. Ed.* **2021**, 60, 20915–20920.
- (5) Giri, N.; Del Pópolo, M. G.; Melaugh, G.; Greenaway, R. L.; Rätzke, K.; Koschine, T.; Pison, L.; Gomes, M. F. C.; Cooper, A. I.; James, S. L. Liquids with permanent porosity. *Nature* **2015**, 527, 216–220.
- (6) Zhang, J. S.; Chai, S. H.; Qiao, Z. A.; Mahurin, S. M.; Chen, J. H.; Fang, Y. X.; Wan, S.; Nelson, K.; Zhang, P. F.; Dai, S. Porous Liquids: A Promising Class of Media for Gas Separation. *Angew. Chem.-Int. Ed.* **2015**, 54, 932–936.
- (7) Li, P.; Chen, H.; Schott, J. A.; Li, B.; Zheng, Y.; Mahurin, S. M.; Jiang, D.-e.; Cui, G.; Hu, X.; Wang, Y.; Li, L.; Dai, S. Porous liquid zeolites: hydrogen bonding-stabilized H-ZSM-5 in branched ionic liquids. *Nanoscale* **2019**, 11, 1515–1519.
- (8) Zhao, X.; Yuan, Y.; Li, P.; Song, Z.; Ma, C.; Pan, D.; Wu, S.; Ding, T.; Guo, Z.; Wang, N. A polyether amine modified metal organic framework enhanced the CO₂ adsorption capacity of room temperature porous liquids. *Chem. Commun.* **2019**, 55, 13179–13182.
- (9) Mow, R. E.; Lipton, A. S.; Shulda, S.; Gaulding, E. A.; Gennett, T.; Braunecker, W. A. Colloidal three-dimensional covalent organic frameworks and their application as porous liquids. *J. Mater. Chem. A* **2020**, 8, 23455–23462.
- (10) Shan, W.; Fulyio, P. F.; Kong, L. Y.; Schott, J. A.; Do-Thanh, C. L.; Tian, T.; Hu, X. X.; Mahurin, S. M.; Xing, H. B.; Dai, S. New Class of Type III Porous Liquids: A Promising Platform for Rational Adjustment of Gas Sorption Behavior. *ACS Appl. Mater. Interfaces* **2018**, 10, 32–36.
- (11) Xin, Y. Y.; Ning, H. L.; Wang, D. C.; Li, X. Q.; Fan, W. D.; Ju, X. Q.; Wang, H. N.; Zhang, Y. C.; Yang, Z. Y.; Yao, D. D.; Zheng, Y. P. A generalizable strategy based on the rule of "like dissolves like" to construct porous liquids with low viscosity for CO capture. *Nano Res.* **2023**, 16, 10369–10380.
- (12) Xin, Y. Y.; Wang, D. C.; Yao, D. D.; Ning, H. L.; Li, X. Q.; Ju, X. Q.; Zhang, Y. C.; Yang, Z. Y.; Xu, Y. H.; Zheng, Y. P. Post-synthetic modification of UiO-66-OH toward porous liquids for CO capture. *New J. Chem.* **2022**, 46, 2189–2197.
- (13) He, S.; Chen, L.; Cui, J.; Yuan, B.; Wang, H.; Wang, F.; Yu, Y.; Lee, Y.; Li, T. General Way To Construct Micro- and Mesoporous Metal-Organic Framework-Based Porous Liquids. *J. Am. Chem. Soc.* **2019**, 141, 19708–19714.
- (14) Greenaway, R. L.; Holden, D.; Eden, E. G. B.; Stephenson, A.; Yong, C. W.; Bennison, M. J.; Hasell, T.; Briggs, M. E.; James, S. L.; Cooper, A. I. Understanding gas capacity, guest selectivity, and diffusion in porous liquids. *Chem. Sci.* **2017**, 8, 2640–2651.
- (15) Liu, S. J.; Liu, J. D.; Hou, X. D.; Xu, T. T.; Tong, J.; Zhang, J. X.; Ye, B. J.; Liu, B. Porous Liquid: A Stable ZIF-8 Colloid in Ionic Liquid with Permanent Porosity. *Langmuir* **2018**, 34, 3654–3660.
- (16) Gabelman, A.; Hwang, S. T. Hollow fiber membrane contactors. *J. Membr. Sci.* **1999**, 159, 61–106.
- (17) Bachman, J. E.; Smith, Z. P.; Li, T.; Xu, T.; Long, J. R. Enhanced ethylene separation and plasticization resistance in polymer membranes incorporating metal-organic framework nanocrystals. *Nat. Mater.* **2016**, 15, 845–849.
- (18) Low, Z. X.; Budd, P. M.; McKeown, N. B.; Patterson, D. A. Gas Permeation Properties, Physical Aging, and Its Mitigation in High Free Volume Glassy Polymers. *Chem. Rev.* **2018**, 118, 5871–5911.
- (19) Yang, X. C.; Ullah, Z.; Stoddart, J. F.; Yavuz, C. T. Porous Organic Cages. *Chem. Rev.* **2023**, 123, 4602–4634.
- (20) Liu, J.; Li, L.; Zhao, Z. T.; Yu, F. Q. Porous Organic Cage Membranes for Biofuel Purification via Pervaporation: A Molecular Simulation Study. *ACS Sustainable Chem. Eng.* **2021**, 9, 14890–14899.
- (21) Kearsley, R. J.; Alston, B.; Briggs, M. E.; Greenaway, R. L.; Cooper, A. I. Accelerated robotic discovery of type II porous liquids. *Chem. Sci.* **2019**, 10, 9454–9465.
- (22) Krishnan, K.; Crawford, J. M.; Thallapally, P. K.; Carreon, M. A. Porous Organic Cages CC3 and CC2 as Adsorbents for the Separation of Carbon Dioxide from Nitrogen and Hydrogen. *Ind. Eng. Chem. Res.* **2022**, 61, 10547–10553.
- (23) Li, Z. J.; Srebnik, S. Expanding carbon capture capacity: uncovering additional CO₂ adsorption sites in imine-linked porous organic cages. *Phys. Chem. Chem. Phys.* **2021**, 23, 10311–10320.
- (24) Liu, C.; Li, W. L.; Liu, Y.; Wang, H. L.; Yu, B. Q.; Bao, Z. B.; Jiang, J. Z. Porous organic cages for efficient gas selective separation and iodine capture. *Chem. Eng. J.* **2022**, 428, No. 131129.
- (25) Slater, A. G.; Little, M. A.; Briggs, M. E.; Jelfs, K. E.; Cooper, A. I. A solution-processable dissymmetric porous organic cage. *Mol. Syst. Des. Eng.* **2018**, 3, 223–227.
- (26) Demir, H.; Aksu, G. O.; Gulbalkan, H. C.; Keskin, S. MOF Membranes for CO₂ Capture: Past, Present and Future. *Carbon Capture Sci. Technol.* **2022**, 2, No. 100026.
- (27) Trickett, C. A.; Helal, A.; Al-Maythaly, B. A.; Yamani, Z. H.; Cordova, K. E.; Yaghi, O. M. The chemistry of metal-organic frameworks for CO₂ capture, regeneration and conversion. *Nat. Rev. Mater.* **2017**, 2, 17045.
- (28) Parkes, M. V.; Greathouse, J. A.; Hart, D. B.; Sava Gallis, D. F.; Nenoff, T. M. Ab initio molecular dynamics determination of competitive O₂ vs. N₂ adsorption at open metal sites of M₂(dobdc). *Phys. Chem. Chem. Phys.* **2016**, 18, 11528–11538.
- (29) Shang, S. S.; Tao, Z. Y.; Yang, C.; Hanif, A.; Li, L. C.; Tsang, D. C. W.; Gu, Q. F.; Shang, J. Facile synthesis of CuBTC and its graphene oxide composites as efficient adsorbents for CO₂ capture. *Chem. Eng. J.* **2020**, 393, No. 124666.
- (30) Gong, X.; Wang, Y. J.; Kuang, T. R. ZIF-8-Based Membranes for Carbon Dioxide Capture and Separation. *ACS Sustainable Chem. Eng.* **2017**, 5, 11204–11214.
- (31) Ji, Y. Z.; Liu, X. Y.; Li, H. C.; Jiao, X.; Yu, X. Q.; Zhang, Y. F. Hydrophobic ZIF-8 covered active carbon for CO₂ capture from humid gas. *J. Ind. Eng. Chem.* **2023**, 121, 331–337.
- (32) Yang, F.; Ge, T. S.; Zhu, X. C.; Wu, J. Y.; Wang, R. Z. Study on CO₂ capture in humid flue gas using amine-modified ZIF-8. *Sep. Purif. Technol.* **2022**, 287, No. 120535.
- (33) Slater, A. G.; Cooper, A. I. Function-led design of new porous materials. *Science* **2015**, 348, No. aaa8075.
- (34) Cho, M.; Lee, S.; Choi, M.; Lee, J. W. Novel Spray Tower for CO₂ Capture Using Uniform Spray of Monosized Absorbent Droplets. *Ind. Eng. Chem. Res.* **2018**, 57, 3065–3075.
- (35) Bon, V.; Kavooosi, N.; Senkovska, I.; Kaskel, S. Tolerance of Flexible MOFs toward Repeated Adsorption Stress. *ACS Appl. Mater. Interfaces* **2015**, 7, 22292–22300.
- (36) Deng, Y. M.; Dewil, R.; Appels, L.; Ansart, R.; Baeyens, J.; Kang, Q. Reviewing the thermo-chemical recycling of waste polyurethane foam. *J. Environ. Manage.* **2021**, 278, No. 111527.
- (37) Rimsza, J. M.; Nenoff, T. M. Porous Liquids: Computational Design for Targeted Gas Adsorption. *ACS Appl. Mater. Interfaces* **2022**, 14, 18005–18015.
- (38) Shi, T.; Zheng, Y. P.; Wang, T. Y.; Li, P. P.; Wang, Y. D.; Yao, D. D. Effect of Pore Size on the Carbon Dioxide Adsorption Behavior of Porous Liquids Based on Hollow Silica. *ChemPhysChem* **2018**, 19, 130–137.
- (39) Wang, D. C.; Xin, Y. Y.; Li, X. Q.; Ning, H. L.; Wang, Y. D.; Yao, D. D.; Zheng, Y. P.; Meng, Z. Y.; Yang, Z. Y.; Pan, Y. T.; Li, P. P.; Wang, H. N.; He, Z. J.; Fan, W. D. Transforming Metal-Organic Frameworks into Porous Liquids via a Covalent Linkage Strategy for CO₂ Capture. *ACS Appl. Mater. Interfaces* **2021**, 13, 2600–2609.

- (40) He, C.; Zou, Y. H.; Si, D. H.; Chen, Z. A.; Liu, T. F.; Cao, R.; Huang, Y. B. A porous metal-organic cage liquid for sustainable CO₂ conversion reactions. *Nat. Commun.* **2023**, *14*, No. 3317.
- (41) Wang, D. C.; Xin, Y. Y.; Li, X. Q.; Wang, F.; Wang, Y. D.; Zhang, W. R.; Zheng, Y. P.; Yao, D. D.; Yang, Z. Y.; Lei, X. F. A universal approach to turn UiO-66 into type 1 porous liquids via post-synthetic modification with corona-canopy species for CO₂ capture. *Chem. Eng. J.* **2021**, *416*, 127625 DOI: 10.1016/j.cej.2020.127625.
- (42) Yin, Z. J.; Chen, H. Y.; Yang, L.; Peng, C. J.; Qin, Y. H.; Wang, T. L.; Sun, W.; Wang, C. W. Investigations of CO₂ Capture from Gas Mixtures Using Porous Liquids. *Langmuir* **2021**, *37*, 1255–1266.
- (43) Ji, D. L.; Xue, C. L.; Wen, Y. T.; Cheng, D.; Luo, H. H.; Li, Y. Microscopic Study on the Performance Optimization of Porous Ionic Liquids for CO₂ Capture by Selection of Crown Ether Solvents. *ACS Sustainable Chem. Eng.* **2022**, *10*, 15263–15272.
- (44) Dinker, M. K.; Li, M. M.; Zhao, K.; Zuo, M. R.; Ding, L. F.; Liu, X. Q.; Sun, L. B. Transformation of Type III to Type II Porous Liquids by Tuning Surface Rigidity of Rhodium(II)-Based Metal-Organic Polyhedra for CO₂ Cycloaddition. *Angew. Chem.-Int. Ed.* **2023**, *62*, No. e202306495.
- (45) Bennett, T. D.; Coudert, F. X.; James, S. L.; Cooper, A. I. The changing state of porous materials. *Nat. Mater.* **2021**, *20*, 1179–1187.
- (46) Wang, D. C.; Ying, Y. P.; Xin, Y. Y.; Li, P. P.; Yang, Z. Y.; Zheng, Y. P. Porous Liquids Open New Horizons: Synthesis, Applications, and Prospects. *Acc. Mater. Res.* **2023**, *4*, 854–866.
- (47) Gui, B.; Lin, G. Q.; Ding, H. M.; Gao, C.; Mal, A.; Wang, C. Three-Dimensional Covalent Organic Frameworks: From Topology Design to Applications. *Acc. Chem. Res.* **2020**, *53*, 2225–2234.
- (48) Nguyen, T. S.; Yavuz, C. T. Quantifying the nitrogen effect on CO₂ capture using isoporous network polymers. *Chem. Commun.* **2020**, *56*, 4273–4275.
- (49) Beuerle, F.; Gole, B. Covalent Organic Frameworks and Cage Compounds: Design and Applications of Polymeric and Discrete Organic Scaffolds. *Angew. Chem.-Int. Ed.* **2018**, *57*, 4850–4878.
- (50) Chang, C. W.; Borne, I.; Lawler, R. M.; Yu, Z. Z.; Jang, S. S.; Lively, R. P.; Sholl, D. S. Accelerating Solvent Selection for Type II Porous Liquids. *J. Am. Chem. Soc.* **2022**, *144*, 4071–4079.
- (51) Gaillac, R.; Pullumbi, P.; Beyer, K. A.; Chapman, K. W.; Keen, D. A.; Bennett, T. D.; Coudert, F. X. Liquid metal-organic frameworks. *Nat. Mater.* **2017**, *16*, 1149–1154.
- (52) Greenaway, R. L.; Santolini, V.; Pulido, A.; Little, M. A.; Alston, B.; Briggs, M. E.; Day, G. M.; Cooper, A. I.; Jelfs, K. E. From Concept to Crystals via Prediction: Multi-Component Organic Cage Pots by Social Self-Sorting. *Angew. Chem.-Int. Ed.* **2019**, *58*, 16275–16281.
- (53) Kai, A. T.; Egleston, B. D.; Tarzia, A.; Clowes, R.; Briggs, M. E.; Jelfs, K. E.; Cooper, A. I.; Greenaway, R. L. Modular Type III Porous Liquids Based on Porous Organic Cage Microparticles. *Adv. Funct. Mater.* **2021**, *31*, No. 2106116.
- (54) Xu, L. J.; Li, M. Z.; Lu, W. Y. Time- and Pressure-Dependent Gas Diffusion in a Nanoconfined Liquid Phase. *J. Phys. Chem. C* **2021**, *125*, 5596–5601.
- (55) Park, S.; Moilanen, D. E.; Fayer, M. D. Water dynamics - The effects of ions and nanoconfinement. *J. Phys. Chem. B* **2008**, *112*, 5279–5290.
- (56) Ivanov, M. Y.; Poryvaev, A. S.; Polyukhov, D. M.; Prikhod'ko, S. A.; Adonin, N. Y.; Fedin, M. V. Nanoconfinement effects on structural anomalies in imidazolium ionic liquids. *Nanoscale* **2020**, *12*, 23480–23487.
- (57) Wang, J. M.; Wolf, R. M.; Caldwell, J. W.; Kollman, P. A.; Case, D. A. Development and testing of a general amber force field. *J. Comput. Chem.* **2004**, *25*, 1157–1174.
- (58) Sprenger, K. G.; Jaeger, V. W.; Pfaendtner, J. The General AMBER Force Field (GAFF) Can Accurately Predict Thermodynamic and Transport Properties of Many Ionic Liquids. *J. Phys. Chem. B* **2015**, *119*, 5882–5895.
- (59) Cornell, W. D.; Cieplak, P.; Bayly, C. I.; Kollman, P. A. Application of RESP charges to calculate conformational energies, hydrogen-bond energies, and free-energies of solvation. *J. Am. Chem. Soc.* **1993**, *115*, 9620–9631.
- (60) Wang, J. M.; Wang, W.; Kollman, P. A.; Case, D. A. Automatic atom type and bond type perception in molecular mechanical calculations. *J. Mol. Graphics Modell.* **2006**, *25*, 247–260.
- (61) Harris, J. G.; Yung, K. H. Carbon Dioxides Liquid-Vapor Coexistence Curve And Critical Properties as Predicted by a Simple Molecular-Model. *J. Phys. Chem. A* **1995**, *99*, 12021–12024.
- (62) Martin, M. G.; Siepmann, J. I. Transferable Potentials for Phase Equilibria. 1. United-atom description of *n*-alkanes. *J. Phys. Chem. B* **1998**, *102*, 2569–2577.
- (63) Potoff, J. J.; Siepmann, J. I. Vapor-liquid equilibria of mixtures containing alkanes, carbon dioxide, and nitrogen. *AIChE J.* **2001**, *47*, 1676–1682.
- (64) Frisch, M. J.; Trucks, G. W.; Schlegel, H. B.; Scuseria, G. E.; Robb, M. A.; Cheeseman, J. R.; Scalmani, G.; Barone, V.; Petersson, G. A.; Nakatsuji, H. et al. *Gaussian 16*. Revision C.01, Wallingford, CT, 2016.
- (65) Slater, A. G.; Reiss, P. S.; Pulido, A.; Little, M. A.; Holden, D. L.; Chen, L. J.; Chong, S. Y.; Alston, B. M.; Clowes, R.; Haranczyk, M.; Briggs, M. E.; Hasell, T.; Day, G. M.; Cooper, A. I. Computationally-Guided Synthetic Control over Pore Size in Isostructural Porous Organic Cages. *ACS Cent. Sci.* **2017**, *3*, 734–742.
- (66) Martínez, L.; Andrade, R.; Birgin, E. G.; Martínez, J. M. PACKMOL: A Package for Building Initial Configurations for Molecular Dynamics Simulations. *J. Comput. Chem.* **2009**, *30*, 2157–2164.
- (67) Martínez, J. M.; Martínez, L. Packing optimization for automated generation of complex system's initial configurations for molecular dynamics and docking. *J. Comput. Chem.* **2003**, *24*, 819–825.
- (68) Essmann, U.; Perera, L.; Berkowitz, M. L.; Darden, T.; Lee, H.; Pedersen, L. G. A smooth particle mesh ewald method. *J. Chem. Phys.* **1995**, *103*, 8577–8593.
- (69) Hess, B.; Bekker, H.; Berendsen, H. J. C.; Fraaije, J. LINCS: A linear constraint solver for molecular simulations. *J. Comput. Chem.* **1997**, *18*, 1463–1472.
- (70) Evans, D. J.; Holian, B. L. The Nose-Hoover Thermostat. *J. Chem. Phys.* **1985**, *83*, 4069–4074.
- (71) Van der Spoel, D.; Lindahl, E.; Hess, B.; Groenhof, G.; Mark, A. E.; Berendsen, H. J. C. GROMACS: Fast, flexible, and free. *J. Comput. Chem.* **2005**, *26*, 1701–1718.
- (72) Huang, X. H.; Margulis, C. J.; Li, Y. H.; Berne, B. J. Why is the partial molar volume of CO₂ so small when dissolved in a room temperature ionic liquid? Structure and dynamics of CO₂ dissolved in [BMIM]⁺[PF6][−]. *J. Am. Chem. Soc.* **2005**, *127*, 17842–17851.
- (73) Avila, J.; Cervinka, C.; Dugas, P. Y.; Pádua, A. A. H.; Gomes, M. C. Porous Ionic Liquids: Structure, Stability, and Gas Absorption Mechanisms. *Adv. Mater. Interfaces* **2021**, *8*, No. 2001982.